

Chapter 4

Continuous Time Fourier Analysis

CHAPTER HIGHLIGHTS

- Introduction
- Approximation of a Signal by Other Signals
- Fourier Series Representation of Periodic Signals
- Properties of Continuous Time Fourier Series
- Continuous Time (C.T.) Fourier Transform
- Relation between Fourier Transform and Laplace transform
- Properties of the Continuous-time Fourier Transform
- Common Fourier Transform Pairs
- Frequency Response of Continuous-time LTI Systems
- LTI Systems Characterized by Differential Equations
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INTRODUCTION

There are four distinct Fourier representations, each applicable to different class of signals. The Fourier series (FS) applies to continuous-time periodic signals, and the discrete-time Fourier series (DTFS) applies to discrete-time periodic signals. Non-periodic signals have Fourier transform representations, The Fourier transform (FT) applies to a signal that is continuous in time and non-periodic. The discrete-time Fourier transform (DTFT) applies to a signal that is discrete in time and non-periodic.

Relationship between time properties of a signal and the appropriate Fourier representation

Time/property	Periodic	Non-periodic
Continuous (t)	Fourier series (FS)	Fourier transform (FT)
Discrete (n)	Discrete-time Fourier series (DTFS)	Discrete-time Fourier transform (DTFT)

APPROXIMATION OF A SIGNAL BY OTHER SIGNALS

Let us consider two signals $f_1(t)$ and $f_2(t)$. We can approximate $f_1(t)$ in terms of $f_2(t)$ over a certain interval ($t_1 < t < t_2$) as $f_1(t) \approx C_{12} f_2(t)$ for ($t_1 < t < t_2$)

We must choose C_{12} such that the error between actual function and the approximated function is minimum and is given as follows:

$$C_{12} = \frac{\int_{t_1}^{t_2} f_1(t) \cdot f_2(t) dt}{\int_{t_1}^{t_2} f_2^2(t) dt}$$

$f_1(t)$ has a component of wave form $f_2(t)$, and this component has a magnitude C_{12} . If C_{12} vanishes, then the signal $f_1(t)$ contains no component of signal $f_2(t)$ and the functions are orthogonal to each other over interval (t_1, t_2) if $\int_{t_1}^{t_2} f_1(t) \cdot f_2(t) dt = 0$,

Similarly, in case of complex functions $f_1(t)$, $f_2(t)$ $f_1(t) \approx C_{12} f_2(t)$ is approximation of $f_1(t)$ in terms of $f_2(t)$, and C_{12} to minimize mean square error magnitude is given by:

$$C_{12} = \frac{\int_{t_1}^{t_2} f_1(t) f_2^*(t) dt}{\int_{t_1}^{t_2} f_2(t) f_2^*(t) dt},$$

Where $f_2^*(t)$ is complex conjugate of $f_2(t)$.

For a set of complex functions $\{g_r(t)\}$ $r = 1, 2, \dots$ mutually orthogonal over the interval (t_1, t_2)

$$\int_{t_1}^{t_2} g_m g_n^*(t) dt = \begin{cases} 0, & \text{if } m \neq n \\ K_m, & \text{if } m = n \end{cases}$$

If this set of functions is complete, then any function $f(t)$ can be expressed as follows:

$$f(t) = C_1 g_1(t) + C_2 g_2(t) + \dots + C_r g_r(t) + \dots$$

Where $C_r = \frac{1}{K_r} \int_{t_1}^{t_2} f(t) g_r^*(t) dt$

$$K_r = \int_{t_1}^{t_2} g_r(t) g_r^*(t) dt$$

Let us consider a set of n functions $g_1(t), g_2(t) \dots g_n(t)$ which are orthogonal to one another over an interval t_1 to t_2 .

$$\int_{t_1}^{t_2} g_j(t) g_k(t) dt = 0, \text{ for } j \neq k \text{ and consider } \int_{t_1}^{t_2} g_j^2(t) dt = K_j$$

an arbitrary function $f(t)$ can be approximated over an interval (t_1, t_2) by a linear combination of these n mutually orthogonal functions

$$f(t) \simeq c_1 g_1(t) + c_2 g_2(t) + \dots + c_k g_k(t) + \dots + c_n g_n(t)$$

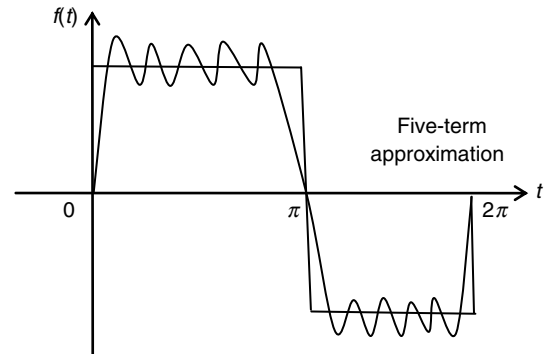
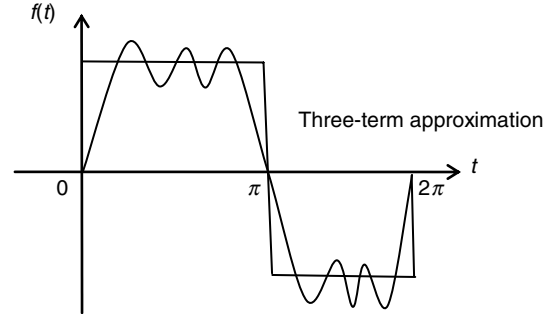
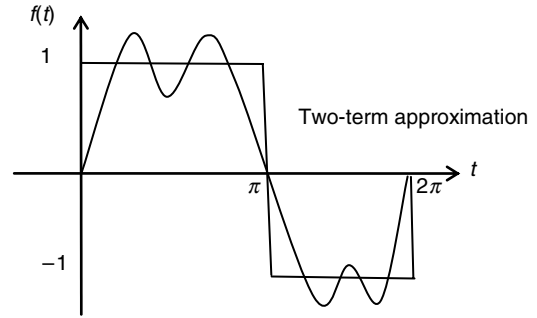
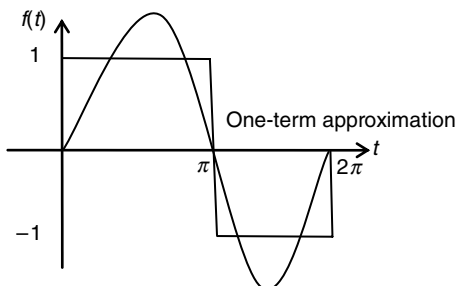
$$= \sum_{r=1}^n c_r g_r(t)$$

For the best approximation, mean square error over the interval has to be minimized then

$$C_r = \frac{\int_{t_1}^{t_2} f(t) g_r(t) dt}{\int_{t_1}^{t_2} g_r^2(t) dt} = \frac{1}{K_r} \int_{t_1}^{t_2} f(t) g_r(t) dt$$

From the above discussion, we can conclude that $f(t)$ contains a component of signal $g_r(t)$, and this component has a magnitude C_r . Representation of $f(t)$ by a set of infinite mutually orthogonal functions is called generalized Fourier series representation of $f(t)$.

The square waves are represented by sinusoids.



From the approximation by sinusoids, the square wave can be obtained by increasing the number of terms.

Some examples of orthogonal functions:

$\sin \omega_0 t, \sin 2\omega_0 t, \sin 3\omega_0 t, \dots$ etc., form orthogonal set over any interval $(t_0, t_0 + 2\pi/\omega_0)$

Similarly, $\cos \omega_0 t, \cos 2\omega_0 t, \cos 3\omega_0 t, \dots$ etc., also form orthogonal set, and in the same interval, similarly, $\cos n\omega_0 t$ and $\sin m\omega_0 t$ are also orthogonal over the same interval.

$$\int_{t_0}^{t_0 + 2\pi/\omega_0} \sin n\omega_0 t \cdot \sin m\omega_0 t dt = 0$$

$$\int_{t_0}^{t_0 + 2\pi/\omega_0} \cos n\omega_0 t \cdot \cos m\omega_0 t dt = 0 \text{ for all } m \neq n$$

$$\int_{t_0}^{t_0 + 2\pi/\omega_0} \sin m\omega_0 t \cdot \cos n\omega_0 t dt = 0$$

So, composite set of functions consisting of set $\cos n\omega_0 t$ and $\sin n\omega_0 t$ for $(n = 0, 1, 2, \dots)$ forms a complete orthogonal set.

$$\int_{t_0}^{t_0+2\pi/\omega_0} \sin^2 n\omega_0 t \cdot dt = \int_{t_0}^{t_0+2\pi/\omega_0} \cos^2 n\omega_0 t \cdot dt = \frac{\pi}{\omega_0} = \frac{T_0}{2}$$

We can represent an arbitrary function $f(t)$ by linear combination of sine and cosine functions which is called as trigonometric Fourier series. A set of exponential functions $\{e^{jn\omega_0 t}\}$ ($n = 0, \pm 1, \pm 2, \pm 3, \dots$) is orthogonal over an interval $(t_0, t_0 + 2\pi/\omega_0)$ for any value of

$$\int_{t_0}^{t_0+2\pi/\omega_0} (e^{jn\omega_0 t})(e^{jm\omega_0 t})^* dt = \frac{2\pi}{\omega_0} = T_0 \text{ for } m = n$$

$$= 0 \text{ for } m \neq n$$

So, we can represent an arbitrary function $f(t)$ by a linear combination of exponential functions over an interval $(t_0, t_0 + 2\pi/\omega_0)$

$$f(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t} \quad (t_0 < t < t_0 + 2\pi/\omega_0)$$

This is called as complex exponential Fourier series representation

$$C_k = \frac{1}{T_0} \int_{t_0}^{t_0+2\pi/\omega_0} f(t) e^{-jk\omega_0 t} dt$$

FOURIER SERIES REPRESENTATION OF PERIODIC SIGNALS

If $x(t) = x(t + T_0)$, then $x(t)$ said to be periodic, the smallest positive value of T_0 is called fundamental period of $x(t)$.

$$x(t) = \cos(\omega_0 t + \phi),$$

$$x(t + T_0) = \cos[\omega_0(t + T_0) + \phi]$$

$$= \cos(\omega_0 t + \phi + \omega_0 T_0)$$

$x(t + T_0) = x(t)$ for minimum value $\omega_0 T_0 = 2\pi$ or for all multiples of 2π , but we need to consider smallest value as fundamental period T_0 .

$$T_0 = \frac{2\pi}{\omega_0} \text{ is fundamental period}$$

$x(t) = e^{j\omega_0 t}$ are periodic where $\omega_0 = \frac{2\pi}{T_0} = 2\pi f_0$ is the fundamental angular frequency.

Consider representing a periodic signal as a weighted super position of complex sinusoidal since the weighted super position must have the same period as the signal, each sinusoid in the superposition must have the same period as the signal, This implies that the frequency of each sinusoid must be an integer multiple of signal's fundamental frequency.

A sinusoid whose frequency is an integer multiple of a fundamental frequency is said to be a harmonic of the sinusoid at the fundamental frequency.

For example, $e^{jk\omega_0 t}$ is the k^{th} harmonic of the $e^{j\omega_0 t}$.

Complex Fourier Series

The complex Fourier series representation of a periodic signal $x(t)$ with fundamental period T_0 and fundamental frequency ω_0 is given by

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}, \quad \omega_0 = \frac{2\pi}{T_0}$$

C_K – complex Fourier coefficient

$$C_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

$\int_{T_0}$ denotes integral over one period of T_0

$C_0 = \frac{1}{T_0} \int_{T_0} x(t) dt$ which indicates C_0 equals the average value of $x(t)$ over one period.

The Fourier series coefficients C_k are known as a frequency-domain representation of $x(t)$ because each Fourier series coefficient is associated with a complex sinusoid of a different frequency.

If $x(t)$ is real, then $C_{-K} = C_K^*$

Trigonometric Fourier Series

The trigonometric Fourier series representation of a periodic signal $x(t)$ with fundamental period T_0 , (or) fundamental angular frequency ω_0 is given by

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos k\omega_0 t + b_k \sin k\omega_0 t)$$

$$\omega_0 = \frac{2\pi}{T_0}$$

$$a_k = \frac{2}{T_0} \int_{T_0} x(t) \cos k\omega_0 t \cdot dt$$

$$b_k = \frac{2}{T_0} \int_{T_0} x(t) \sin k\omega_0 t \cdot dt$$

The trigonometric Fourier coefficient a_k , b_k , and complex Fourier coefficient C_K are related as

$$\frac{a_0}{2} = c_0, \quad a_k = c_k + c_{-k}, \quad b_k = j(c_k - c_{-k})$$

$$c_k = \frac{1}{2}(a_k - jb_k), \quad c_{-k} = \frac{1}{2}(a_k + jb_k)$$

Even and Odd Signals

If a periodic signal $x(t)$ is even, then $b_k = 0$, and its Fourier series contains only cosine terms.

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos k\omega_0 t, \quad \omega_0 = \frac{2\pi}{T_0}$$

If $x(t)$ is odd, then $a_k = 0$, and its Fourier series contains only sine terms

$$x(t) = \sum_{k=-\infty}^{\infty} b_k \sin k\omega_0 t, \quad \omega_0 = \frac{2\pi}{T_0}$$

Harmonic Form of Fourier Series

Other form of the Fourier series representation of a real periodic signal $x(t)$ with fundamental period T_0 is given by [known as harmonic form Fourier series of $x(t)$]

$$x(t) = D_0 + \sum_{k=1}^{\infty} D_k \cos(k\omega_0 t - \theta_k), \quad \omega_0 = \frac{2\pi}{T_0}$$

The term D_0 is known as d.c. component, and the term $D_k \cos(k\omega_0 t - \theta_k)$ is referred as the k^{th} harmonic component of $x(t)$.

The coefficients D_k and θ_k are related to trigonometric Fourier series coefficients a_k and b_k as follows:

$$D_0 = \frac{a_0}{2}, \quad D_k = \sqrt{a_k^2 + b_k^2}, \quad \theta_k = \tan^{-1} \left(\frac{b_k}{a_k} \right)$$

Dirichlet Conditions

For the convergence of Fourier series

- (1) $x(t)$ is absolutely integrable over any period

$$\int_{T_0} |x(t)| dt < \infty$$

- (2) $x(t)$ has a finite number of maxima and minima with in any field interval of ' T '.
- (3) $x(t)$ has a finite number of discontinuities with in any finite interval of ' T ', and each of these discontinuities is finite.

Effects of Symmetry on Fourier Coefficients

Symmetry	Condition	The trigonometric Fourier coefficient to be zero
Even	$x(t) = x(-t)$	$b_n = 0$, no sine terms will be present
Odd	$x(-t) = -x(t)$	$a_n = 0$, no cosine terms will be present
Half wave	$x(t) = -x(t \pm \frac{T}{2})$	d.c. term $a_0 = 0$; a_n , $b_n = 0$ for n even. All even coefficients will be zero

Amplitude and Phase Spectra of Periodic Signal

Complex Fourier coefficients C_k can be expressed as $C_k = |C_k| e^{j\phi_k}$

A plot $|C_k|$ versus angular frequency ω is called amplitude spectrum of $x(t)$.

A plot of ϕ_k versus ω is called the phase spectrum of $x(t)$.

Both are discrete in nature.

For a real, periodic signal $x(t)$ we have,

$$|C_{-k}| = |C_k|, \quad \phi_{-k} = -\phi_k$$

Therefore, amplitude spectrum is an even function of ω , whereas phase spectrum is an odd function of ω , for real periodic signal.

PROPERTIES OF CONTINUOUS TIME FOURIER SERIES

$x(t)$ is a periodic signal with period T_0 and fundamental frequency $\omega_0 = 2\pi/T_0$, then the Fourier series coefficients of $x(t)$ are denoted by C_k , then we can use the notation $x(t) \xrightarrow{\text{FS}} C_k$

1. Linearity

If $x(t)$ and $y(t)$ denote two periodic signals with period T_0 and which have Fourier coefficients denoted by a_k and b_k , respectively,

$$x(t) \xrightarrow{\text{FS}} a_k, \quad y(t) \xrightarrow{\text{FS}} b_k$$

as $x(t)$, and $y(t)$ have the same period T_0 , it easily follows that any linear combination of the two signals will also be periodic with period T_0 .

$z(t) = Ax(t) + By(t)$, then Fourier series coefficients of $z(t)$ are $C_k = Aa_k + Bb_k$

$$z(t) = Ax(t) + By(t) \xrightarrow{\text{FS}} C_k = Aa_k + Bb_k$$

2. Time shifting

If $x(t) \xrightarrow{\text{FS}} a_k$, then

$$x(t - t_0) \xrightarrow{\text{FS}} e^{-jk\omega_0 t_0} a_k = e^{-jk(2\pi/T_0)t_0} a_k$$

When a periodic signal is shifted in time, the magnitude of its Fourier series coefficients remain unaltered, only phase will be changed.

3. Time reversal

If $x(t) \xrightarrow{\text{FS}} C_k$, then $x(-t) \xrightarrow{\text{FS}} C_{-k}$

In other words, time reversal applied to a continuous time signal results in a time reversal of the corresponding sequence of Fourier series coefficients. If $x(t)$ is even, then $x(-t) = x(t)$, then its Fourier series coefficients are also even $C_{-k} = C_k$. Similarly, if $x(t)$ is odd, then $x(-t) = -x(t)$, then so are its Fourier series coefficients $C_{-k} = -C_k$.

4. Frequency shifting

If $x(t) \xrightarrow{\text{FS}} C_k$, then $e^{jm\omega_0 t} x(t) \xrightarrow{\text{FS}} C_{k-m}$

5. Time scaling

If $x(t)$ is periodic with period T_0 , and fundamental frequency $\omega_0 = 2\pi/T_0$, then $x(at)$, $a > 0$, is periodic with period $\frac{T_0}{a}$ and fundamental frequency $a\omega_0$, then

Fourier series representation of

$$x(at) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

The Fourier coefficients C_k have not changed, but the Fourier series representation has changed because of the change in fundamental frequency.

6. Multiplication

If $x(t) \xrightarrow{\text{FS}} a_k$, $y(t) \xrightarrow{\text{FS}} b_k$, and $x(t)$, $y(t)$ are periodic with period T_0 , then $x(t) \cdot y(t)$ is also periodic with period T_0 , then

$$x(t) \cdot y(t) \xrightarrow{\text{FS}} C_k = \sum_{\ell=-\infty}^{\infty} a_{\ell} b_{k-\ell}$$

C_K can be interpreted as the discrete-time convolution of the sequence representing Fourier series coefficients of $x(t)$ and $y(t)$

7. Conjugation and conjugate symmetry

If $x(t)$ FS C_k , then $x^*(t)$ FS C_{-k}^*

If $x(t)$ is real signal, then $x(t) = x^*(t)$

Therefore, $C_k = C_{-k}^*$ (or) $C_k^* = C_{-k}$

Hence, we can write $|C_k| = |C_{-k}|$, $\angle C_k = -\angle C_{-k}$

$\text{Re}\{C_k\} = \text{Re}\{C_{-k}\}$, $\text{Im}\{C_k\} = -\text{Im}\{C_{-k}\}$

When $x(t)$ is real $C_k^* = C_{-k}$

When $x(t)$ is even, then $C_{-k} = C_k$

Therefore, when $x(t)$ is real and even $C_k = C_k^*$ and $C_k = C_{-k}$, that is, the Fourier series coefficients are also real and even.

When $x(t)$ is real, $C_k^* = C_{-k}$

and when $x(t)$ is odd, $C_{-k} = -C_k$

By combining both, $C_k^* = -C_{-k}$

So, when $x(t)$ is real and odd, the Fourier series coefficients are also odd and purely imaginary ($C_k^* = -C_{-k}$)

8. Periodic convolution

If $x(t)$ and $y(t)$ are periodic with period T_0 , then $x(t)$ FS a_k , $y(t)$ FS b_k

$$z(t) = \int_{T_0} x(\tau)y(t-\tau)d\tau \quad \text{FS } C_k = T_0 a_k b_k$$

9. Differentiation

If $x(t)$ FS C_k is periodic with angular frequency ω_0 ,

then $\frac{dx(t)}{dt}$ FS $jk\omega_0 C_k$.

10. Integration

If $x(t)$ is finite valued, periodic (ω_0), and if $C_0 = 0$,

$$\text{then } \int_{-\infty}^t x(t) dt \quad \text{FS } \frac{1}{jk\omega_0} C_k.$$

11. Even odd decomposition of real signals

If $x(t)$ can be written as $x(t) = x_e(t) + x_o(t)$ as sum of even and odd components of $x(t)$.

$$x(-t) = x_e(t) - x_o(t)$$

$$x_e(t) = \frac{x(t) + x(-t)}{2}, \quad x_o(t) = \frac{x(t) - x(-t)}{2}$$

When $x(t)$ is real, $x(t)$ FS C_k , $x(-t)$ FS C_{-k}

Similarly, $C_k = C_{-k}^*$ (or) $C_{-k} = C_k^*$

$$\begin{aligned} \text{So, for } x_e(t) &= \frac{1}{2} \{x(t) + x(-t)\} \quad \text{FS } \frac{1}{2} \{C_k + C_{-k}\} \\ &= \text{Re}\{C_k\} \end{aligned}$$

$$\begin{aligned} \text{for } x_o(t) &= \frac{1}{2} \{x(t) - x(-t)\} \quad \text{FS } \frac{1}{2} \{C_k - C_{-k}\} \\ &= j\text{Im}\{C_k\} \end{aligned}$$

$$x_e(t) \quad \text{FS } \text{Re}\{C_k\}$$

$$x_o(t) \quad \text{FS } j\text{Im}\{C_k\}$$

12. Parseval's theorem

The average power of periodic signal $x(t)$ over any period is

$$P = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |C_k|^2$$

Parseval's theorem states that the total average power in a periodic signal equals the sum of the average powers in all of its harmonic components

Solved Examples

Example 1

Which of the following signal will not have Fourier series representation?

(A) $\sin 4t + \cos 9t$

(B) $1 + \cos 2\pi t$

(C) e^{-j5t}

(D) $\cos 4\pi t + \sin 3t$

Solution

Only periodic signals will have Fourier series representation

For $\sin 4t + \cos 9t \rightarrow$ period is LCM of $\left(\frac{2\pi}{4}, \frac{2\pi}{9}\right) = 2\pi$

$\cos 2\pi t$ is periodic with period $T_0 = \frac{2\pi}{2\pi} = 1$, and D.C.

component 1 is present in the signal $1 + \cos 2\pi t$ is also periodic.

For e^{-j5t} periodic with period $\frac{2\pi}{5}$.

However, $\cos 4\pi t + \sin 3t$ is not periodic, $T_1 = \frac{2\pi}{4\pi} = \frac{1}{2}$,

$T_2 = \frac{2\pi}{3}$, $\frac{T_1}{T_2}$ is not rational, so option (D) will not have

Fourier series expansion.

CONTINUOUS TIME (C.T.) FOURIER TRANSFORM

The function $X(\omega)$ and $x(t)$ are said to be Fourier transform pairs defined as follows:

$$x(t) \leftrightarrow X(\omega)$$

The Fourier transform of $x(t)$ is

$$X(\omega) = F\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

The inverse Fourier transform of $X(\omega)$ is

$$x(t) = F^{-1}\{X(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Fourier Spectra

Fourier transform of $x(t)$ can be expressed as $X(\omega) = |X(\omega)| e^{j\phi(\omega)}$

$X(\omega)$, Fourier transform of non-periodic signal $x(t)$ is the frequency-domain specification of $x(t)$ and is referred as Fourier spectrum of $x(t)$,

$|X(\omega)|$ is called magnitude spectrum of $x(t)$

$\phi(\omega)$ is called the phase spectrum of $x(t)$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

For real signal $x(t)$

$$X(-\omega) = \int_{-\infty}^{\infty} x(t) e^{j\omega t} dt$$

So, $X(-\omega) = X^*(\omega)$

And $|X(\omega)| = |X(-\omega)|$, $\phi(-\omega) = -\phi(\omega)$

Here, the amplitude spectrum $|X(\omega)|$ is an even function and the phase spectrum $\phi(\omega)$ is an odd function of ω .

Convergence of Fourier Transform

(Dirichlet conditions)

- (1) $x(t)$ is absolutely integrable $\int_{-\infty}^{\infty} |x(t)| dt < \infty$
- (2) $x(t)$ has finite number of maxima and minima with in any finite interval
- (3) $x(t)$ has a finite number of discontinuities with in any finite interval, and each of these discontinuities is finite.

RELATION BETWEEN FOURIER TRANSFORM AND LAPLACE TRANSFORM

Fourier transform of $x(t)$ is $X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$ The

bilateral Laplace transform of $x(t)$ is $X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$

We can observe from these equations, Fourier transform is special case of Laplace transform in which $s = j\omega$

i.e., $|X(s)|_{s=j\omega} = F\{x(t)\}$

If we substitute $s = \sigma + j\omega$ in Laplace transform equation

$$\begin{aligned} X(\sigma + j\omega) &= \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt \\ &= \int_{-\infty}^{\infty} (x(t) e^{-\sigma t}) e^{-j\omega t} dt = F\{x(t) e^{-\sigma t}\} \end{aligned}$$

Bilateral Laplace transform of $x(t)$ can be interpreted as the Fourier transform of $x(t) e^{-\sigma t}$. If $x(t)$ is absolutely integrable, the Fourier transform of $x(t)$ can be obtained from the Laplace transform of $x(t)$ with $S = j\omega$.

Example 2

Find the Fourier transform of the function $f(t) = e^{-at} u(t)$.

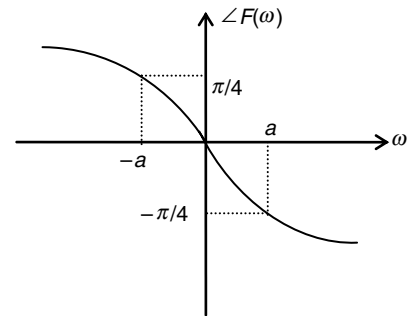
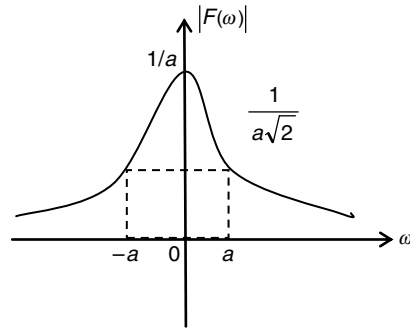
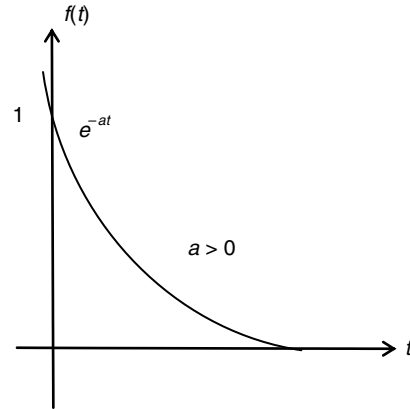
Solution

The Fourier transform does not converge for $a \leq 0$, since $\int_{-\infty}^{\infty} e^{-at} u(t) dt = \infty$ for $a \leq 0$. $f(t)$ is not absolutely integrable for $a \leq 0$.

For $a > 0$,

$$F(\omega) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-(a+j\omega)t} dt$$

$$\begin{aligned} &= \frac{-1}{a+j\omega} e^{(-a+j\omega)t} \Big|_0^{\infty} \\ &= \frac{1}{a+j\omega} = \frac{-1}{\sqrt{a^2+\omega^2}} \angle -\tan^{-1}(\omega/a) \end{aligned}$$



Example 3

Find the Fourier transform of the rectangular pulse

$$x(t) = 1 \quad |t| < T_0$$

$$0 \quad |t| > T_0$$

Solution

$x(t)$ is absolutely integrable for $T_0 < \infty$

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \\ &= \int_{-T_0}^{T_0} 1 \cdot e^{-j\omega t} dt \end{aligned}$$

$$= \frac{-1}{j\omega} e^{-j\omega t} \Big|_{-T_0}^{T_0}$$

$$= \frac{-1}{j\omega} (e^{-j\omega T_0} - e^{+j\omega T_0}) = 2 \frac{\sin \omega T_0}{\omega}$$

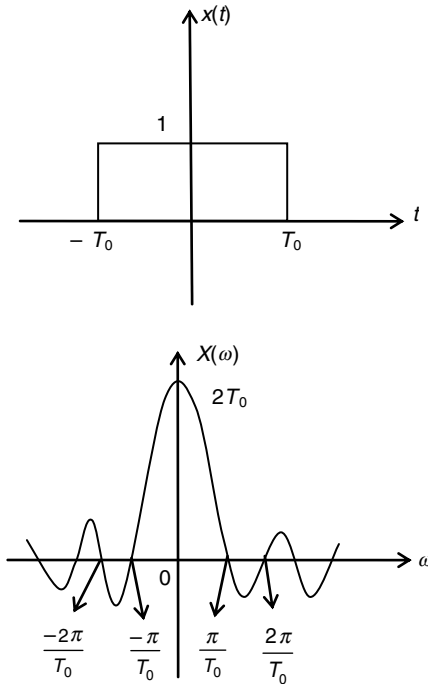
Sinc function can be defined as follows:

$$\text{Sinc}(x) = \frac{\sin \pi x}{\pi x}$$

$$X(\omega) = 2T_0 \frac{\sin \omega T_0}{\omega T_0} = 2T_0 \text{Sinc}(\omega T_0/\pi)$$

when $\omega \rightarrow 0$, $X(\omega) = \lim_{\omega \rightarrow 0} \frac{2T_0 \sin \omega T_0}{\omega T_0}$

$$= 2T_0 \text{ (by using L Hospital's rule)}$$



The magnitude spectrum is $|X(\omega)| = 2 \left| \frac{\sin(\omega T_0)}{\omega} \right|$

Phase spectrum is $\angle X(\omega) = 0$, $\begin{cases} \sin(\omega T_0)/\omega > 0 \\ \pi, \sin \omega T_0/\omega < 0 \end{cases}$

From the above example, we can see that if T_0 increases, then non-zero extent of $x(t)$ increases, and $X(\omega)$ becomes more concentrated about the frequency origin. Similarly, when T_0 decreases $x(t)$ width reduces, $X(\omega)$ becomes less concentrated about the frequency origin. The duration of $x(t)$ is inversely related to the width or bandwidth of $X(\omega)$.

Example 4

Find the Fourier transform of the $f(t) = \delta(t)$.

Solution

$$F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = e^{j\omega \cdot 0} = 1$$

$$\delta(t) \xrightarrow{\text{FT}} 1$$

Time-domain signal defined at origin, but frequency components are present from $\omega = -\infty$ to ∞ .

Example 5

What is the inverse Fourier transform of $F(\omega) = 2\pi\delta(\omega)$?

Solution

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi \delta(\omega) e^{j\omega t} d\omega = \frac{1}{2\pi} 2\pi e^{j\omega t} = 1$$

So, inverse Fourier transform of $2\pi \delta(\omega)$ is 1

$$1 \leftrightarrow 2\pi\delta(\omega)$$

This implies that the frequency content of a d.c. signal is concentrated entirely at $\omega = 0$,

PROPERTIES OF THE CONTINUOUS TIME FOURIER TRANSFORM

1. Linearity

$$a_1 x_1(t) + a_2 x_2(t) \leftrightarrow a_1 X_1(\omega) + a_2 X_2(\omega)$$

2. Time shifting

$$x(t - t_0) \leftrightarrow e^{-j\omega t_0} X(\omega)$$

By delaying a signal by t_0 seconds $x(t - t_0)$, the magnitude spectrum $|X(\omega)|$ will not be changed, but the phase spectrum $\angle X(\omega)$ will be changed by $-\omega t_0$. So, time delay in a signal causes a linear phase shift in its spectrum.

3. Frequency shifting

$$e^{j\omega_0 t} x(t) \leftrightarrow X(\omega - \omega_0)$$

The multiplication of a signal by a factor of $e^{j\omega_0 t}$ shifts the spectrum of that signal by $\omega = \omega_0$. We can observe the duality between the time shifting and frequency shifting properties.

The multiplication of a signal $x(t)$ by a sinusoid of frequency ω_0 shifts the spectrum $X(\omega)$ by $\pm \omega_0$

$$x(t) \cos \omega_0 t = \frac{1}{2} [x(t) e^{j\omega_0 t} + x(t) e^{-j\omega_0 t}] \leftrightarrow \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$$

4. Time scaling

$$x(at) \leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

The scaling property states that time compression of a signal results in its spectral expansion, and time expansion of the signal results in its spectral compression.

The scaling property implies that if $x(t)$ is wider, its spectrum is narrower, and vice versa, doubling the signal duration halves its bandwidth, and vice versa. This suggests that the band width of a signal is inversely proportional to the signal duration or width (in seconds)

5. Time reversal

$$x(-t) \leftrightarrow X(-\omega)$$

For time scaling property, if we consider $a = -1$, we can obtain the inversion property of time and frequency

6. Duality $X(t) \leftrightarrow 2\pi x(-\omega)$

7. Differentiation in the time domain

$$\frac{d}{dt} x(t) \leftrightarrow j\omega X(\omega)$$

$$\frac{d^n}{dt^n} x(t) \leftrightarrow (j\omega)^n X(\omega)$$

8. Differentiation in the frequency domain

$$(-jt) x(t) \leftrightarrow \frac{d}{d\omega} X(\omega)$$

Differentiation in time domain corresponds to multiplication by $j\omega$ in the frequency domain

9. Integration in time domain

$$\int_{-\infty}^t x(\tau) d\tau \leftrightarrow \pi X(0) \delta(\omega) + \frac{1}{j\omega} X(\omega)$$

Integration in the time domain corresponds to division by $j\omega$ in the frequency domain and the impulse term reflects the d.c. or average value that can result from integration.

10. Convolution

$$x_1(t) * x_2(t) \leftrightarrow X_1(\omega) X_2(\omega)$$

The Fourier transform maps the convolution of two signals into the product of their Fourier transforms.

11. Multiplication

$$x_1(t) x_2(t) \leftrightarrow \frac{1}{2\pi} X_1(\omega) * X_2(\omega)$$

Multiplication in the time domain corresponds to convolution in frequency domain. We can observe the duality property between time and frequency domains from above two properties.

12. Parseval's relations

$$\int_{-\infty}^{\infty} x_1(\lambda) X_2(\lambda) d\lambda = \int_{-\infty}^{\infty} X_1(\lambda) x_2(\lambda) d\lambda$$

$$\int_{-\infty}^{\infty} x_1(t) x_2(t) dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\omega) X_2(-\omega) d\omega$$

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

Parseval's identity says that energy content E of $x(t)$, can be computed by integrating $|X(\omega)|^2$ over all frequencies, for this reason $|X(\omega)|^2$ is often called as energy density spectrum of $x(t)$.

13. Real signal

If $x(t)$ is real, and let $x(t) = x_e(t) + x_o(t)$

Where $x_e(t)$, $x_o(t)$ are even and odd components of $x(t)$,

Fourier transform of $x(t)$ is $X(\omega) = A(\omega) + jB(\omega)$ then $X(-\omega) = X^*(\omega)$

So, we can conclude that

$$x_e(t) \xrightarrow{\text{FT}} \text{Re} \{X(\omega)\} = A(\omega)$$

$$x_o(t) \xrightarrow{\text{FT}} j \text{Im} \{X(\omega)\} = jB(\omega)$$

This shows that the Fourier transform of an even signal is a real function of ω ,

And Fourier transform of an odd signal is a pure imaginary function of ω .

The $X(\omega)$ is complex conjugate symmetric for real signal $x(t)$.

When $X^*(\omega) = X(-\omega)$

By taking the real and imaginary parts of this expression gives $\text{Re} \{X(\omega)\} = \text{Re} \{X(-\omega)\}$ and $\text{Im} \{X(\omega)\} = -\text{Im} \{X(-\omega)\}$

In other words, if $x(t)$ is real valued, then the real part of the transform is an even function of frequency, while the imaginary part is an odd function of frequency.

This also implies that the magnitude spectrum is an even function of frequency, while the phase spectrum is an odd function.

COMMON FOURIER TRANSFORM PAIRS

$x(t)$	$X(\omega)$
$\delta(t)$	1
$\delta(t - t_0)$	$e^{-j\omega t_0}$
1	$2\pi\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos \omega_0 t$	$\pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin \omega_0 t$	$-j\pi[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
$u(t)$	$\pi\delta(\omega) + \frac{1}{j\omega}$
$u(-t)$	$\pi\delta(\omega) - \frac{1}{j\omega}$
$e^{-at} u(t), a > 0$	$\frac{1}{j\omega + a}$
$t e^{-at} u(t), a > 0$	$\frac{1}{(j\omega + a)^2}$
$e^{-a t }, a > 0$	$\frac{2a}{a^2 + \omega^2}$

(Continued)

$x(t)$	$X(\omega)$
$\frac{1}{a^2 + t^2}$	$e^{-a \omega }$
$e^{-at^2}, a > 0$	$\sqrt{\frac{\pi}{a}} e^{-\omega^2/4a}$
$\rho_a(t) = \begin{cases} 1 & t < a \\ 0 & t > a \end{cases}$	$2 \frac{\sin \omega a}{\omega}$
$\frac{\sin at}{\pi t}$	$\rho_a(\omega) = \begin{cases} 1 & \omega < a \\ 0 & \omega > a \end{cases}$
$\text{sgn}(t)$	$\frac{2}{j\omega}$
$\sum_{k=-\infty}^{\infty} \delta(t - kT)$	$\omega_0 \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0), \omega_0 = \frac{2\pi}{T}$

Fourier Transform Representation of Periodic Signals

If $x(t)$ is a periodic signal with fundamental frequency ω_0 , then the Fourier series representation of $x(t)$ is

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} dt, \omega_0 = \frac{2\pi}{T_0}$$

where C_k – complex Fourier coefficient

$$C_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-jk\omega_0 t} dt$$

We have 1 FT $2\pi \delta(\omega)$

By using frequency shifting property $e^{jk\omega_0 t}$ FT $2\pi \delta(\omega - k\omega_0)$

We can apply linearity property of the Fourier transform to obtain

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t} \xrightarrow{\text{FT}} X(\omega) = 2\pi \sum_{k=-\infty}^{\infty} c_k \delta(\omega - k\omega_0)$$

Thus, the Fourier transform of a periodic signal is a series of impulses spaced by the fundamental frequency ω_0 . The k^{th} impulse has strength of $2\pi C_k$, C_k is k^{th} Fourier series coefficient.

Fourier Transform in Terms of Frequency (f)

We can change the variable ω to f (in Hertz), by writing $\omega = 2\pi f$ and $d\omega = 2\pi df$

$$\text{So, } X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

$$\text{and } x(t) = \int_{-\infty}^{\infty} X(2f) e^{j2\pi ft} df$$

We can get $X(f)$ by changing ω to $2\pi f$.

$$\text{Similarly, } \delta(\omega) = \delta(2\pi f) = \frac{1}{2\pi} \delta(f)$$

$$x(t) = e^{-at} u(t) \rightarrow X(\omega) = \frac{1}{a + j\omega} \rightarrow X(f) = \frac{1}{a + j2\pi f}$$

$$x(t) = e^{-|t|} \rightarrow X(\omega) = \frac{2a}{a^2 + \omega^2} \rightarrow X(f) = \frac{2a}{a^2 + 4\pi^2 f^2}$$

$$x(t) = u(t) \rightarrow X(\omega) = \pi \delta(\omega) + \frac{1}{j\omega} \rightarrow X(f) = \frac{1}{2} \delta(f) + \frac{1}{j2\pi f}$$

$$x(t) = e^{-pt^2} \rightarrow X(\omega) = e^{-\omega^2/4\pi} \rightarrow x(f) = e^{-\pi f^2}$$

$$\cos(2\pi at) \rightarrow \pi [\delta(\omega + 2\pi a) + \delta(\omega - 2\pi a)] \rightarrow$$

$$\frac{1}{2} [\delta(f + a) + \delta(f - a)]$$

$$\sin(2\pi at) \rightarrow j\pi [\delta(\omega + 2\pi a) - \delta(\omega - 2\pi a)] \rightarrow$$

$$\frac{j}{2} [\delta(f + a) - \delta(f - a)]$$

Example 6

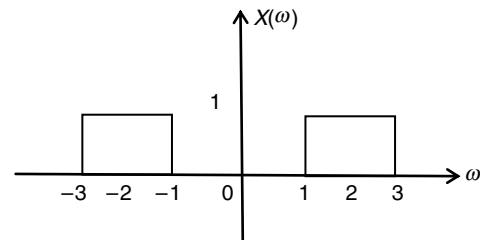
Find the Fourier transform of $x(t) = e^{3t} u(-t)$.

Solution

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = \int_{-\infty}^0 e^{3t} e^{-j\omega t} dt \\ &= \int_0^{\infty} e^{(3-j\omega)t} dt \text{ replace } t \text{ with } -t \\ &= -\int_0^{\infty} e^{-(3-j\omega)t} dt = \frac{-e^{-(3-j\omega)t}}{(3-j\omega)} = \frac{1}{3-j\omega} \end{aligned}$$

Example 7

Find the inverse Fourier transform for the spectra depicted in figure.



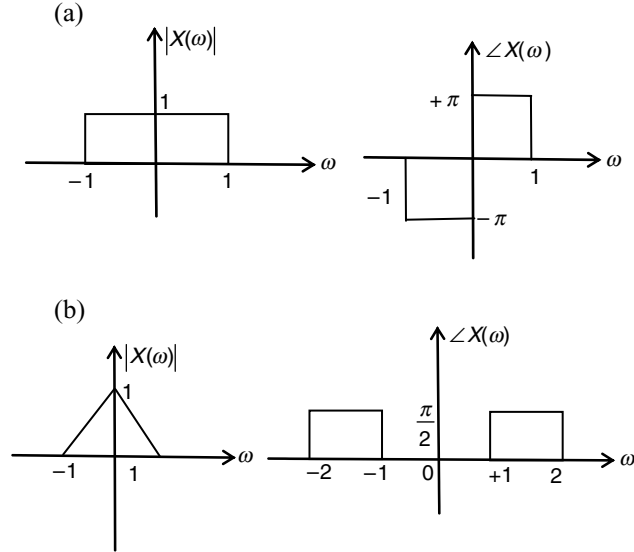
Solution

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \\ \text{So, } x(t) &= \frac{1}{2\pi} \left[\int_{-3}^{-1} 1 e^{j\omega t} d\omega + \int_1^3 1 e^{j\omega t} d\omega \right] \\ &= \frac{1}{2\pi} \left\{ \frac{e^{j\omega t}}{jt} \Big|_{-3}^{-1} + \frac{e^{j\omega t}}{jt} \Big|_1^3 \right\} \\ &= \frac{1}{2\pi jt} (e^{-jt} - e^{-j3t} + e^{j3t} - e^{jt}) \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{\pi t} \left\{ \left(\frac{e^{j3t} - e^{-j3t}}{2j} \right) - \left(\frac{e^{jt} - e^{-jt}}{2j} \right) \right\} \\
 &= \frac{1}{\pi t} (\sin 3t - \sin t)
 \end{aligned}$$

Example 8

For the frequency-domain signal depicted in figure, find the nature of time-domain signal?



Solution

For (a), $|X(\omega)|$ is an even function and $\angle X(\omega)$ is an odd function.

So, we can say that $x(t)$ is real and odd.

For (b), $|X(\omega)|$ is an even function as well as $\angle X(\omega)$ is also an even function so $x(t)$ is real and even.

Example 9

Find $x(t)$ for the spectra

$$X(\omega) = \begin{cases} j\omega |\omega| < 1 \\ 0 & |\omega| > 1 \end{cases}$$

Solution

$$\begin{aligned}
 X(\omega) &= j\omega |\omega| < 1 \\
 0 &|\omega| > 1
 \end{aligned}$$

We can see $X(\omega) = j\omega z(\omega)$

Where $Z(\omega) = 1 \quad |\omega| < 1$

$$0 \quad |\omega| > 1$$

Then we can have $z(t) = \frac{1}{\pi t} \sin t$

$$z(t) \xrightarrow{\text{FT}} Z(\omega)$$

$$\text{then } \frac{dz(t)}{dt} \leftrightarrow j\omega Z(\omega) = X(\omega)$$

The inverse Fourier transform of $X(\omega)$ is

$$x(t) = \frac{d}{dt} z(t) = \frac{d}{dt} \left(\frac{1}{\pi t} \sin t \right)$$

$$x(t) = \frac{1}{\pi t} \cos t - \frac{1}{\pi t^2} \sin t$$

Example 10

Find the time-domain signal corresponding to the Fourier transform

$$X(\omega) = \frac{5j\omega + 12}{-\omega^2 + 5j\omega + 6}$$

Solution

$$\begin{aligned}
 X(\omega) &= \frac{5(j\omega) + 12}{(j\omega)^2 + 5(j\omega) + 6} \\
 &= \frac{5j\omega + 12}{(j\omega + 2)(j\omega + 3)} = \frac{A}{j\omega + 2} + \frac{B}{j\omega + 3} \\
 &= \frac{5 \times -2 + 12}{-2 + 3} \cdot \frac{1}{j\omega + 2} + \frac{5 \times -3 + 12}{-3 + 2} \cdot \frac{1}{j\omega + 3} = \frac{2}{j\omega + 2} + \frac{3}{j\omega + 3}
 \end{aligned}$$

By taking inverse Fourier transform.

$$x(t) = 2e^{-2t}u(t) + 3e^{-3t}u(t)$$

Example 11

What is the inverse Fourier transform of the step function in frequency? ($X(\omega) = u(\omega)$)

Solution

$X(\omega) = u(\omega)$ from the duality property $x(t) \xrightarrow{\text{FT}} X(\omega)$

$$X(t) \xrightarrow{\text{FT}} 2\pi x(-\omega)$$

We have time reversal property

$$\begin{aligned}
 g(t) &\leftrightarrow G(\omega), \text{ then} \\
 g(-t) &\leftrightarrow G(-\omega)
 \end{aligned}$$

so we can write $X(-t) \leftrightarrow 2\pi x(\omega)$

$$\frac{1}{2\pi} X(-t) \leftrightarrow x(\omega)$$

If we consider $x(t) = u(t)$, then $X(\omega) = \pi \delta(\omega) + \frac{1}{j\omega}$

then inverse Fourier transform of $u(\omega)$ is $\frac{1}{2\pi} X(-t)$

$$= \frac{1}{2\pi} \left\{ \pi \delta(-t) - \frac{1}{jt} \right\} = \frac{\delta(t)}{2} - \frac{1}{2\pi jt} = \frac{\delta(t)}{2} + \frac{j}{2\pi t}$$

As $\delta(-t) = \delta(t)$

Example 12

Find $x(t)$ if $X(\omega) = j \frac{d}{d\omega} \left\{ \frac{e^{j3\omega}}{1 + j\omega/2} \right\}$.

Solution

Consider $p(t) = e^{-t}u(t) \leftrightarrow P(\omega) = \frac{1}{1 + j\omega}$

$$\text{then } X(\omega) = j \frac{d}{d\omega} \{e^{j3\omega} P(\omega/2)\}$$

If $Q(\omega) = P(\omega/2)$, then $q(t) = 2p(2t) = 2e^{-2t} u(2t) = 2e^{-2t} u(t)$
 {time scaling $|a|$ $x(at) \leftrightarrow X(\omega/a)$ }

$$\text{Now } X(\omega) = j \frac{d}{d\omega} \{e^{j3\omega} Q(\omega)\}$$

If $R(\omega) = e^{j3} Q(\omega)$, then $r(t) = q(t+3) = 2e^{-2(t+3)} u(t+3)$
 $u(t+3)$ {time shifting $x(t-t_0) \leftrightarrow e^{-j\omega t_0} X(\omega)$ }

$$\begin{aligned} \text{Now } X(\omega) &= j \frac{d}{d\omega} \{R(\omega)\}, \text{ then } x(t) = t r(t) \\ &= 2t e^{-2(t+3)} u(t+3) \end{aligned}$$

[differentiation in frequency domain

$$tx(t) \leftrightarrow j \frac{d}{d\omega} \{x(\omega)\}].$$

Example 13

Find the Fourier transform of the impulse train $x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_0)$.

Solution

The fundamental period of $x(t)$ is T_0 , so $\omega_0 = \frac{2\pi}{T_0}$ is fundamental frequency.

The Fourier coefficients are

$$C_k = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \delta(t) e^{-jk\omega_0 t} dt = \frac{1}{T_0}$$

The Fourier transform of a periodic signal is

$$\begin{aligned} X(\omega) &= 2\pi \sum_{k=-\infty}^{\infty} c_k \delta(\omega - k\omega_0) = 2\pi \sum_{k=-\infty}^{\infty} \frac{1}{T_0} \delta(\omega - k\omega_0) \\ &= \frac{2\pi}{T_0} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0) \end{aligned}$$

Hence, Fourier transform of an impulse train is also an impulse train.

FREQUENCY RESPONSE OF CONTINUOUS TIME LTI SYSTEMS

Consider a continuous time LTI system with impulse response $h(t)$, and the input is $x(t) = e^{j\omega t}$, then the convolution integral gives the output as

$$\begin{aligned} y(t) &= \int_{-\infty}^{\infty} h(\tau) e^{j\omega(t-\tau)} d\tau = \int_{-\infty}^{\infty} h(\tau) e^{j\omega t} \cdot e^{-j\omega\tau} d\tau \\ &= e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau = H(\omega) e^{j\omega t} \end{aligned}$$

$$\text{Where we can define } H(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau.$$

The output of the system is, thus, a complex sinusoid of the same frequency as the input, multiplied by the complex number $H(\omega)$.

$H(\omega)$ is a function of only the frequency ω , and not the time t , and is termed the frequency response of the continuous time system.

Output $y(t)$ of a continuous time LTI system. Equals the convolution of the input $x(t)$, with the impulse response $h(t)$.

$$y(t) = x(t) \times h(t) \text{ taking Fourier transform}$$

$$Y(\omega) = X(\omega) H(\omega)$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

$H(\omega)$ is called frequency response of the system.

$$H(\omega) = |H(\omega)| e^{j\theta_H(\omega)}$$

$|H(\omega)|$ is called magnitude response of the system. $\theta_H(\omega)$ is phase response of the system.

The behaviour of a continuous time LTI system in the frequency domain is completely characterized by its frequency response $H(\omega)$

$$\text{Input } X(\omega) = |X(\omega)| e^{j\theta_x(\omega)}$$

$$H(\omega) = |H(\omega)| e^{j\theta_H(\omega)}$$

$$\text{And output } Y(\omega) = |Y(\omega)| e^{j\theta_y(\omega)}$$

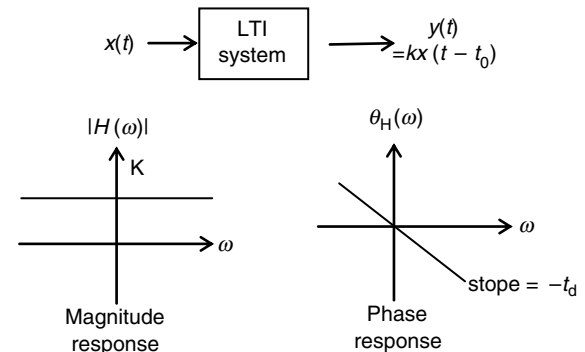
$$\text{Then } Y(\omega) = X(\omega) \cdot H(\omega)$$

$$\text{So, } |Y(\omega)| = |X(\omega)| \cdot |H(\omega)|$$

$$\theta_y(\omega) = \theta_x(\omega) + \theta_H(\omega).$$

For an LTI system, with real-valued impulse response $h(t)$ and frequency response $H(\omega)$, [i.e., Fourier transform of $h(t)$], the output $y(t)$ for an input $x(t) = A \cos(\omega t - \phi)$ is $y(t) = |H(\omega)| A \cos(\omega t - \phi + \angle\{H(\omega)\})$ the system modifies the amplitude of input sinusoid by $H(\omega)$ and the phase by $\angle\{H(\omega)\}$

Distortionless Transmission



For distortionless transmission through LTI system we require, the exact input signal to be reproduced at the output, even its amplitude is different, or it may be time delayed.

If $x(t)$ is input signal, then output $y(t) = k x(t - t_d)$.

By taking Fourier transform $Y(\omega) = k e^{-j\omega t_d} X(\omega)$. So, we can write for distortionless transmission system must have the frequency response.

$$H(\omega) = |H(\omega)| e^{j\theta_H(\omega)} = k e^{-j\omega t_d}$$

$$|H(\omega)| = k, \theta_H(\omega) = -j\omega t_d$$

That is, amplitude of $H(\omega)$ must be constant over the entire frequency range and the phase of $H(\omega)$ must be linear with frequency.

Amplitude Distortion and Phase Distortion

When the amplitude spectrum $|H(\omega)|$ of the system is not constant within the frequency band of interest, the frequency components of the input signal are transmitted with different amount of gain or attenuation. This effect is called amplitude distortion.

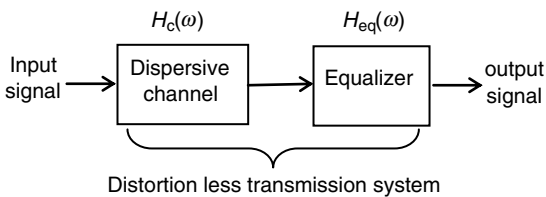
When the phase spectrum $\theta_H(\omega)$ of system is not linear with frequency, the output signal has a different wave form than the input signal because of different delays in passing through the system for different frequency components of the input signals. This form of distortion is called as phase distortion.

Equalization

To compensate for linear distortion, we may use a network known as equalizer, connected in cascade with the system in question.

The equalizer is designed in such a way that, inside the frequency band of interest the overall magnitude and phase responses of this cascade connection approximate the conditions for distortionless transmission to within prescribed limits.

Consider, for example, a communication channel with frequency response $H_C(\omega)$, let an equalizer of frequency response $H_{eq}(\omega)$ be connected in cascade with the channel.



For overall transmission through the cascade connection to be distortionless, we require $H_C(\omega) \cdot H_{eq}(\omega) = e^{-j\omega t_d}$

Where t_d is constant time delay, k is unity here.

Therefore, the frequency response of the equalizer is inversely related to that of the channel, $H_{eq}(\omega) = \frac{e^{-j\omega t_d}}{H_C(\omega)}$

Example 14

Let $x(t) = \frac{1}{\pi t} \sin(2\pi t)$ be the input to a system with impulse response $h(t) = \frac{1}{\pi t} \sin(3\pi t)$, then the output of the system $y(t)$ is?

Solution

$$y(t) = x(t) \times h(t),$$

We can convert to frequency domain by taking Fourier transform $Y(\omega) = X(\omega) H(\omega)$

We have the Fourier transform pair

$$\frac{\sin at}{\pi t} \xrightarrow{\text{FT}} P_a(\omega) = 1 \quad |\omega| < a$$

$$0 \quad |\omega| > a$$

$$x(t) \xrightarrow{\text{FT}} X(\omega) = 1 \quad |\omega| < 2\pi$$

$$0 \quad |\omega| > 2\pi$$

and

$$h(t) \xrightarrow{\text{FT}} H(\omega) = 1 \quad |\omega| < 3\pi$$

$$0 \quad |\omega| > 3\pi$$

then $Y(\omega) = X(\omega) \cdot H(\omega)$

$$0 \quad |\omega| > 2\pi$$

$$\text{Hence, } y(t) = \frac{1}{\pi t} \sin(2\pi t)$$

Example 15

The output of an LTI system in response to an input $x(t) = e^{-t} u(t)$ is $y(t) = e^{-2t} u(t)$. Find the frequency response and the impulse response of the system?

Solution

$$x(t) = e^{-t} u(t) \xrightarrow{\text{FT}} X(\omega) = \frac{1}{j\omega + 1}$$

$$y(t) = e^{-2t} u(t) \xrightarrow{\text{FT}} Y(\omega) = \frac{1}{j\omega + 2}$$

$$y(t) = x(t) * h(t) \xrightarrow{\text{FT}} Y(\omega) = X(\omega) \cdot H(\omega)$$

$$\therefore H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

The frequency response $H(\omega) = \frac{j\omega + 1}{j\omega + 2}$ to find impulse response we need to take inverse Fourier transform for $H(\omega)$

$$H(\omega) = \frac{j\omega + 2 - 1}{j\omega + 2} = 1 - \frac{1}{j\omega + 2}$$

So, $h(t) = \delta(t) - e^{-2t} u(t)$

Example 16

Find the zero state response of a stable LTI system with frequency response $H(\omega) = \frac{1}{j\omega + 3}$ and the input is $x(t) = e^{-2t} u(t)$.

Solution

$$\text{Here } X(\omega) = \frac{1}{j\omega + 2}, H(\omega) = \frac{1}{j\omega + 3}$$

then $Y(\omega) = H(\omega) X(\omega)$

$$= \frac{1}{(j\omega + 2)(j\omega + 3)} = \frac{1}{j\omega + 2} - \frac{1}{j\omega + 3}$$

By taking inverse Fourier transform

$$y(t) = (e^{-2t} - e^{-3t}) u(t)$$

LTI Systems Characterized by Differential Equations

$$\sum_{k=0}^N a_k \frac{d^k}{dt^k} y(t) = \sum_{k=0}^M b_k \frac{d^k}{dt^k} x(t),$$

By taking Fourier transforms on both sides

$$\sum_{k=0}^N a_k (j\omega)^k Y(\omega) = \sum_{k=0}^M b_k (j\omega)^k X(\omega)$$

$$\text{They } H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{\sum_{k=0}^M b_k (j\omega)^k}{\sum_{k=0}^N a_k (j\omega)^k}$$

Which is similar to in Laplace transform if $s = j\omega$.

Phase and Group Delays

Whenever a signal is transmitted through a dispersive (i.e., frequency selective) system, some delay is introduced into the output signal in relation to the input signal. The delay is determined by the phase response of the system. $\phi = \angle\{H(j\omega)\} = \arg\{H(j\omega)\}$

Where $H(j\omega)$ is the frequency response of the channel. Suppose that a sinusoidal signal is transmitted through the channel at a frequency ω_c . The signal received at the channel output lags the transmitted signals by $\phi(\omega_c)$ radians. The time delay corresponding to this phase lag is simply equal

$$\text{to } \tau_p = \frac{-\phi(\omega_c)}{\omega_c}$$

Where $-$ sign accounts for the lag. However, this phase delay is not necessarily the true signals delay, this follows from the fact that a sinusoidal signal has infinite duration, with each cycle exactly like the preceding one. Such a signal does not convey information. When a modulated signal is transmitted through a communication channel, there are two different delays to be considered.

$$(1) \text{ The carrier or phase delay } \tau_p = \frac{-\phi(\omega_c)}{\omega_c}$$

(2) The envelope or group delay

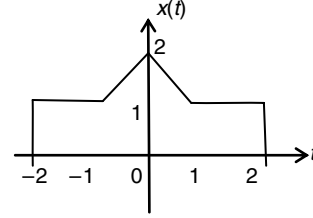
$$\tau_g = \left. \frac{-d\phi(\omega)}{d\omega} \right|_{\omega=\omega_c}$$

The group delay is the true signal delay.

Example 17

For the signal $x(t)$ Shown in figure, find $X(0)$ and

$$\int_{-\infty}^{\infty} X(\omega) d\omega$$



Solution

We know the Fourier transform of $x(t)$ is $X(\omega) =$

$$\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$\text{When } \omega = 0, X(0) = \int_{-\infty}^{\infty} x(t) e^{0} dt = \int_{-\infty}^{\infty} x(t) dt$$

$\int_{-\infty}^{\infty} x(t) dt$ is the area under the curve $x(t)$

$$X(0) = \int_{-\infty}^{\infty} x(t) dt = 4 \times 1 + \frac{1}{2} \times 1 \times 1 \times 2 = 5$$

$$\text{Inverse Fourier transform is } x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$\text{When } t = 0, \int_{-\infty}^{\infty} X(\omega) d\omega = 2\pi x(0) = 2\pi \times 5 = 10\pi$$

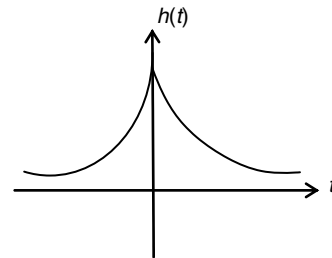
Example 18

Fourier transform of a signal $x(t)$ is given by $X(\omega) = \frac{\omega^2 + 28}{\omega^2 + 16}$, then $x(t)$ is

Solution

$$x(\omega) = \frac{\omega^2 + 28}{\omega^2 + 16} = \frac{\omega^2 + 16 + 12}{\omega^2 + 16} = 1 + \frac{12}{\omega^2 + 16}$$

$$\text{Consider } h(t) = e^{-a|t|} = \begin{cases} e^{-at}; & t > 0 \\ e^{at}; & t < 0 \end{cases}$$



$$h(t) = e^{-at} u(t) + e^{at} u(-t),$$

by taking Fourier transform on both sides

$$H(\omega) = \frac{1}{a + j\omega} + \frac{1}{a - j\omega} = \frac{2a}{a^2 + \omega^2}$$

$$e^{-a}|t| \leftrightarrow \frac{2a}{\omega^2 + a^2}$$

$$\text{Here, } X(\omega) = 1 + \frac{12}{\omega^2 + 16} = 1 + \frac{3}{2} \frac{8}{\omega^2 + 16}$$

by taking inverse Fourier transform,

$$x(t) = \delta(t) + \frac{3}{2} e^{-4}|t|$$

Example 19

Determine the inverse Fourier transform of $X(\omega) = 4\pi \delta(\omega) - 2\pi \delta(\omega - 2\pi) + 2\pi \delta(\omega + 2\pi)$

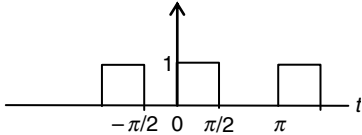
- (A) $2[1 - \sin 2\pi t]$ (B) $2j[1 - \sin 2\pi t]$
 (C) $2[1 - j\sin 2\pi t]$ (D) $2j[1 - j\sin 2\pi t]$

Solution

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} (4\pi \delta(\omega) - 2\pi \delta(\omega - 2\pi) + 2\pi \delta(\omega + 2\pi)) e^{j\omega t} d\omega \\ &= 2 - e^{j2\pi t} + e^{-j2\pi t} \\ &= 2 - 2j \left(\frac{e^{j2\pi t} - e^{-j2\pi t}}{2j} \right) \\ &= 2 - 2j \sin 2\pi t = 2(1 - j \sin 2\pi t) \end{aligned}$$

Example 20

Determine the Fourier series coefficients for periodic signal $x(t)$ as



- (A) $\frac{j}{k\pi} [1 - e^{-jk\pi}]$ (B) $\frac{1}{jk\pi} [1 + e^{-jk\pi}]$
 (C) $\frac{1}{jk\pi} [1 - e^{-jk\pi}]$ (D) $\frac{j}{jk\pi} [1 + e^{-jk\pi}]$

Solution

$$\begin{aligned} c_k &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t} dt, T = \pi, \omega_0 = 2 \\ c_k &= \frac{2}{T} \frac{(e^{-jk\omega_0 t})_0^{\pi/2}}{-jk\omega_0} = \frac{1}{jk\pi} \left(1 - e^{-jk\omega_0 \frac{\pi}{2}} \right) \\ &= \frac{1}{jk\pi} (1 - e^{-jk\pi}) \end{aligned}$$

Example 21

Find the Fourier transform of the signal $x(t) = te^{-2}|t-3|$

- (A) $e^{-3j\omega} \left(\frac{12}{4 + \omega^2} - \frac{j8\omega}{(4 + \omega^2)^2} \right)$
 (B) $e^{3j\omega} \left(\frac{12}{4 + \omega^2} + \frac{j8\omega}{(4 + \omega^2)^2} \right)$
 (C) $e^{-3j\omega} \left(\frac{12}{4 + \omega^2} + \frac{j8\omega}{(4 + \omega^2)^2} \right)$
 (D) $e^{3j\omega} \left(\frac{12}{4 + \omega^2} - \frac{j8\omega}{(4 + \omega^2)^2} \right)$

Solution

$$\begin{aligned} e^{-2}|t| &\rightarrow \frac{4}{4 + \omega^2} \\ x_1(t - 3) &\rightarrow e^{-3j\omega} x_1(\omega) \\ tx_2(t) &\rightarrow \frac{jd}{d\omega} x_2(\omega) \\ X(\omega) &= j \frac{d}{d\omega} \left(e^{-3j\omega} \frac{4}{4 + \omega^2} \right) \\ &= j \frac{d}{d\omega} \left(e^{-3j\omega} \frac{d}{d\omega} \left(\frac{4}{4 + \omega^2} \right) + \left(\frac{4}{4 + \omega^2} \right) \frac{d}{d\omega} (e^{-3j\omega}) \right) \\ &= j \left(\frac{-e^{-3j\omega} 8\omega}{(4 + \omega^2)^2} - 3j \left(\frac{4}{4 + \omega^2} \right) e^{-3j\omega} \right) \\ &= e^{-3j\omega} \left(\frac{12}{4 + \omega^2} - \frac{j8\omega}{(4 + \omega^2)^2} \right) \end{aligned}$$

Example 22

Determine the Inverse Fourier transform of $X(\omega) =$

$$\frac{2j\omega + 5}{(j\omega + 2)^2}$$

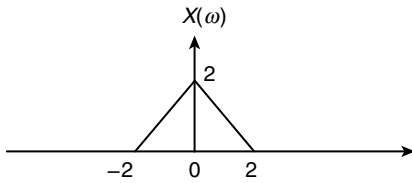
- (A) $(2e^{-2t} - te^{-2t}) u(t)$ (B) $(2e^{-2t} + te^{-2t}) u(t)$
 (C) $(2e^{-2t} - te^{-2t}) u(t)$ (D) $(2e^{-2t} - te^{-2t}) u(t)$

Solution

$$\begin{aligned} X(\omega) &= \frac{2j\omega + 5}{(j\omega + 2)^2} = \frac{2j\omega + 4}{(j\omega + 2)^2} + \frac{1}{(j\omega + 2)^2} \\ &= \frac{2(j\omega + 2)}{(j\omega + 2)^2} + \frac{1}{(j\omega + 2)^2} \\ &= \frac{2}{j\omega + 2} + \frac{1}{(j\omega + 2)^2} \\ x(t) &= 2e^{-2t} u(t) + te^{-2t} u(t) \\ x(t) &= (2e^{-2t} + te^{-2t}) u(t) = (2 + t) e^{-2t} u(t) \end{aligned}$$

Example 23

Consider the signal $X(\omega)$



Find $\int_{-\infty}^{\infty} [x(t)]^2 dt$?

- (A) 0 (B) -1
(C) $2/\pi$ (D) $8/3\pi$

Solution

According to Parseval's

$$\int_{-\infty}^{\infty} x(t)^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} X^2(\omega) d\omega = \frac{1}{2\pi} \cdot \frac{16}{3} = \frac{8}{3\pi}$$

Example 24

A real-valued continuous time signal $x(t)$ has a fundamental period $T = 16$, the non-zero Fourier series coefficients for $x(t)$ are $x(2) = x(-2) = 1$, $x(4) = x(-4) = j$, the signal $x(t)$

would be?

- (A) $\cos \frac{\pi}{4} t + \cos \frac{\pi}{2} t$
(B) $\cos \frac{\pi}{4} t + \cos \left(\frac{\pi}{2} t + \frac{\pi}{2} \right) t$
(C) $2 \left[\cos \frac{\pi}{4} t + \cos \left(\frac{\pi}{2} t + \frac{\pi}{2} \right) \right]$
(D) $2 \left[\cos \frac{\pi}{2} t + \cos \left(\frac{\pi}{4} t + \frac{\pi}{2} \right) \right]$

Solution

$$x(t) = \sum_{k=-\infty}^{\infty} x(k) e^{jk\omega_0 t}$$

$$= x(-2)e^{-j2\omega_0 t} + x(2)e^{j2\omega_0 t} + x(-4)e^{-j4\omega_0 t} + x(4)e^{j4\omega_0 t}$$

$$= e^{-j2\left(\frac{\pi}{8}\right)t} + e^{j2\left(\frac{\pi}{8}\right)t} - e^{-j4\left(\frac{\pi}{8}\right)t} + je^{j4\left(\frac{\pi}{8}\right)t}$$

$$= 2 \left[\frac{e^{j\frac{\pi}{4}t} + e^{-j\frac{\pi}{4}t}}{2} \right] - 2 \left[\frac{e^{j\frac{\pi}{2}t} - e^{-j\frac{\pi}{2}t}}{2j} \right]$$

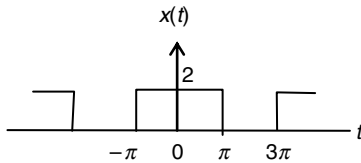
$$= 2 \left[\cos \frac{\pi}{4} t - \sin \frac{\pi}{2} t \right]$$

$$= 2 \left[\cos \frac{\pi}{4} t + \cos \left(\frac{\pi}{2} t + \frac{\pi}{2} \right) \right]$$

EXERCISES**Practice Problems 1**

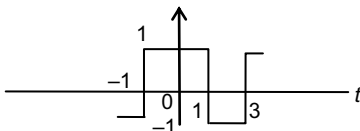
Direction for questions 1 to 25: Select the correct alternative from the given choices.

1. Determine the Fourier series coefficient for periodic signal $x(t)$ as



- (A) $\frac{2}{k\pi} \sin \frac{k\pi}{4}$ (B) $\frac{2}{k\pi} \sin \frac{k\pi}{2}$
(C) $\frac{4}{k\pi} \sin \frac{k\pi}{4}$ (D) $\frac{4}{k\pi} \sin \frac{k\pi}{2}$

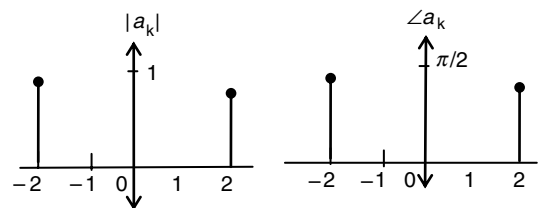
2. Determine the Fourier series coefficient for periodic signal $x(t)$ as



- (A) $\frac{1}{2\pi k} [2e^{-jk\pi/2} - e^{-3jk\pi/2} + e^{jk\pi/2}]$
(B) $\frac{j}{2\pi k} [2e^{jk\pi/2} + e^{-3jk\pi/2} + e^{jk\pi/2}]$
(C) $\frac{j}{2\pi k} [2e^{-jk\pi/2} - e^{-3jk\pi/2} + e^{jk\pi/2}]$
(D) $\frac{1}{2\pi k} [2e^{-jk\pi/2} + e^{-3jk\pi/2} - e^{jk\pi/2}]$

3. Fourier series coefficient of the time domain signal as $\omega_0 = 2\pi$

Determine the corresponding time-domain signal.



- (A) $e^{j\pi/2} \cos 2\pi t$ (B) $2e^{j\pi/2} \cos 4\pi t$
(C) $2e^{-j\pi/2} \cos 2\pi t$ (D) $2e^{-j\pi/2} \cos 4\pi t$

4. Suppose the periodic signal $x(t)$ has fundamental period ' T ' and Fourier coefficients a_k , let b_k be the Fourier coefficients of $y(t)$, where $y(t) = \frac{d^2 x(t)}{dt^2}$, the Fourier coefficient a_k in terms of b_k will be

(A) $\frac{-T^2 b_k}{4\pi^2 k^2}$ (B) $\frac{T^2 b_k}{2\pi^2 k^2}$
 (C) $\frac{T^2 b_k}{\pi^2 k^2}$ (D) $\frac{2T^2 b_k}{\pi^2 k^2}$

5. Find the Fourier transform of signal $x(t) = e^{-3t} u(t-2)$

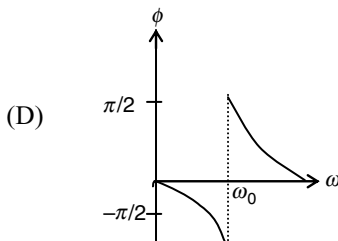
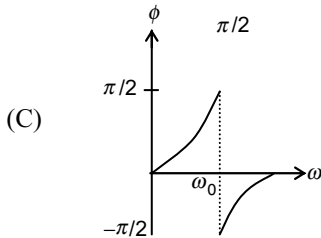
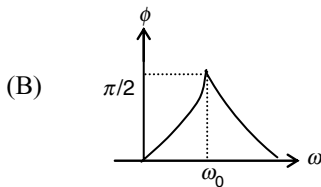
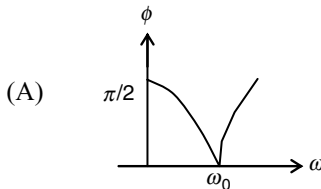
(A) $\frac{e^{-2(3+j\omega)}}{3+j\omega}$ (B) $\frac{e^{-3(2+j\omega)}}{2+j\omega}$
 (C) $\frac{e^{-2(3-j\omega)}}{3-j\omega}$ (D) $\frac{e^{-3(2-j\omega)}}{2-j\omega}$

6. Determine the Inverse Fourier Transform of $X(\omega) = \frac{-j\omega + 6}{\omega^2 - 13j\omega - 36}$

(A) $(3e^{-6t} + 2e^{-4t}) u(t)$ (B) $(3e^{-6t} - 2e^{-4t}) u(t)$
 (C) $(3e^{-9t} - 2e^{-4t}) u(t)$ (D) $(3e^{-9t} + 2e^{-4t}) u(t)$

7. Which one of the following represents the phase response of the function?

$$H(s) = \frac{s^2 + \omega_0^2}{s^2 + \left(\frac{\omega_0}{\theta}\right)s + \omega_0^2}$$



8. Which of the following statements are true?

- (1) A Fourier series for an even periodic function will consist entirely of cosine terms.
 (2) A Fourier series for an odd periodic function will consist entirely of sine terms.
 (3) A Fourier series for an odd periodic function will consist entirely of cosine terms.
 (4) A Fourier series for an even periodic function will consist entirely of sine terms.
 (A) 3, 4 (B) 1, 2 (C) 1, 3 (D) 2, 4

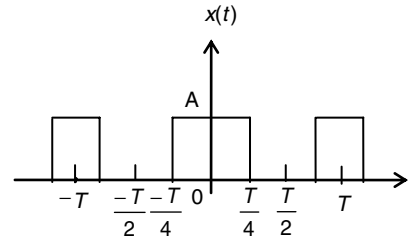
9. The effect of time displacement τ in a periodic function ($n\omega_0$ frequency)

- (A) Frequency spectrum remains same
 (B) Magnetic spectrum remains constant
 (C) Magnetic spectrum shifts by $n\omega_0 t$
 (D) Phase spectrum remains constant

10. Which of the following is true?

- (1) $x(t)$ is real and even function, then the Fourier series coefficients are real and even
 (2) $x(t)$ is real and odd function, then the Fourier series coefficients are imaginary and odd.
 (A) Both (B) Neither (1) nor (2)
 (C) Only (1) (D) Only (2)

11. Fourier series of the wave form shown



(A) $A + \frac{A}{\pi} \left[\cos \omega_0 t - \frac{\cos 3\omega_0 t}{3} + \frac{\cos 5\omega_0 t}{5} \dots \right]$
 (B) $\frac{A}{2} + \frac{2A}{\pi} \left[\cos \omega_0 t - \frac{\cos 3\omega_0 t}{3} + \frac{\cos 5\omega_0 t}{5} \dots \right]$
 (C) $\frac{A}{2} + \frac{2A}{\pi} \left[\cos \omega_0 t - \frac{\cos 4\omega_0 t}{4} + \frac{\cos 6\omega_0 t}{6} \dots \right]$
 (D) $\frac{A}{2} - \frac{2A}{\pi} \left[\cos \omega_0 t - \frac{\cos 3\omega_0 t}{3} + \frac{\cos 5\omega_0 t}{5} \dots \right]$

12. Match the following Fourier transform pairs

Group (1)		Group (2)	
P	$e^{-j\omega t_0}$	1.	$\delta(t)$
Q.	$e^{-a t }$	2.	$\delta(t - t_0)$
R.	1	3.	$\frac{1}{a^2 + t^2}$
S.	$2\pi\delta(\omega)$	4.	1

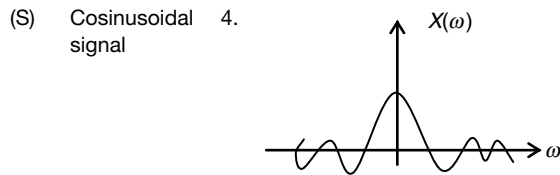
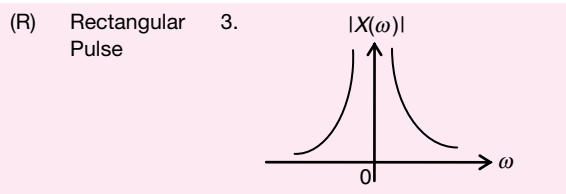
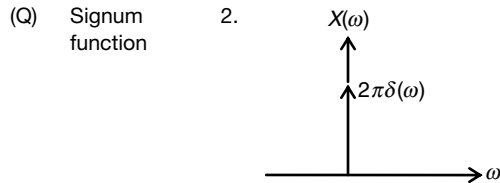
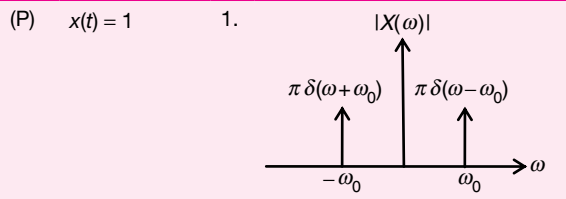
- (A) $P-4, Q-3, R-2, S-1$
 (B) $P-3, Q-2, R-1, S-4$

- (C) $P = 2, Q = 3, R = 1, S = 4$
 (D) $P = 2, Q = 3, R = 4, S = 1$

13. A rectangular function defined as $f(t) = 1/2, 0 < t < \pi, -1/2, (\pi < t < 2\pi)$, then find approximated function by a wave form $\sin t$ over the interval $(0, 2\pi)$ such that the mean square error is minimum.

- (A) $\frac{4}{\pi} \sin t$ (B) $\frac{8}{\pi} \sin t$
 (C) $\frac{2}{\pi} \sin t$ (D) $\frac{1}{\pi} \sin t$

14. Match the following with their magnitude spectrum



- (A) $P = 2, Q = 1, R = 4, S = 3$
 (B) $P = 1, Q = 2, R = 3, S = 4$
 (C) $P = 2, Q = 3, R = 1, S = 4$
 (D) $P = 2, Q = 3, R = 4, S = 1$

15. Find the inverse Fourier transform of $\frac{7 + j\omega}{(3 + j\omega)^2}$

- (A) $(4t - 1)e^{-3t}u(t)$ (B) $(4t + 1)e^{3t}u(t)$
 (C) $(4 + 4)e^{-3t}u(t)$ (D) $(4t + 1)e^{-3t}u(t)$

16. The impulse response of an LTI system is $h(t) = e^{-2t}u(t)$, find the response of the system for $x(t) = e^{-6t}u(t)$

- (A) $\frac{1}{4}(e^{-2t} + e^{-6t})u(t)$ (B) $\frac{1}{4}(e^{-2t} - e^{-6t})u(t)$
 (C) $\frac{-1}{4}(e^{-2t} - e^{-6t})u(t)$ (D) $\frac{-1}{4}(e^{-2t} + e^{-6t})u(t)$

Direction for questions 17 and 18:

17. Fourier transform of the signal $x(t)$ is $X(\omega) = \frac{1}{\omega^2}(e^{j\omega} - j\omega e^{-j\omega} - 1)$. Find the Fourier transform of $x(-t)$

- (A) $\frac{1}{\omega^2}(e^{-j\omega} + j\omega e^{j\omega} - 1)$
 (B) $\frac{1}{\omega^2}(e^{-j\omega} - j\omega e^{j\omega} - 1)$
 (C) $\frac{1}{\omega^2}(e^{j\omega} + j\omega e^{-j\omega} - 1)$
 (D) $\frac{1}{\omega^2}(e^{-j\omega} - j\omega e^{-j\omega} - 1)$

18. Find the Fourier transform of $x(-t - 1)$

- (A) $\frac{e^{j\omega}}{\omega^2}(e^{-j\omega} + j\omega e^{j\omega} - 1)$
 (B) $\frac{e^{-j\omega}}{\omega^2}(e^{-j\omega} + j\omega e^{-j\omega} - 1)$
 (C) $\frac{e^{-j\omega}}{\omega^2}(e^{j\omega} + j\omega e^{j\omega} - 1)$
 (D) $\frac{e^{-j\omega}}{\omega^2}(e^{-j\omega} - j\omega e^{j\omega} - 1)$

19. The Fourier transform of $e^{-t}u(t)$ is $\frac{1}{1 + j\omega}$, then Fourier transform of $\frac{1}{1 + t}$ is

- (A) $2\pi e^{-j\omega}u(j\omega)$ (B) $2\pi e^{j\omega}u(j\omega)$
 (C) $2\pi e^{-j\omega}u(-j\omega)$ (D) $2\pi e^{j\omega}u(-j\omega)$

20. Fourier coefficients of $x(n)$ are

$C_k = \{3, 2 + j, 1, 2 - j\}$. The value of $x(6)$ is

- (A) 1 (B) 3
 (C) $2 - j$ (D) $2 + j$

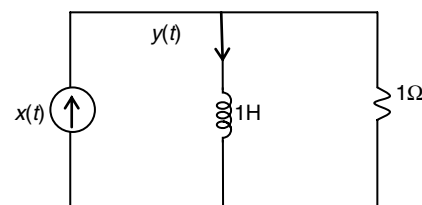
21. Frequency response $H(j\omega)$ of causal LTI system of difference equation

$$y(n) + \frac{1}{3}y[n-1] = x[n]$$

- (A) $\frac{1}{1 + \frac{1}{3}e^{-j\omega}}$ (B) $\frac{1}{1 - \frac{1}{3}e^{-j\omega}}$
 (C) $\frac{1}{1 - \frac{1}{6}e^{-j\omega}}$ (D) $\frac{1}{1 + \frac{1}{4}e^{-j\omega}}$

Direction for questions 22 and 23:

22. Consider a causal LTI system implemented as the RL circuit shown



A current source produces an input current $x(t)$, and system output is considered to be the current $y(t)$ flowing through inductor. Find the differential equation relating to $x(t)$ and $y(t)$

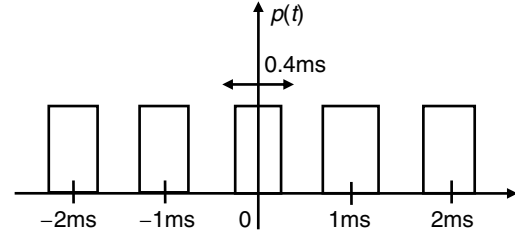
- (A) $\frac{dy(t)}{dt} - y(t) = x(t)$ (B) $\frac{dy(t)}{dt} = x(t)$
 (C) $y(t) = 0$ (D) $\frac{dy(t)}{dt} + y(t) = x(t)$

23. The frequency response of this system for input $x(t) = e^{j\omega t}$

- (A) $H(j\omega) = 1$ (B) $H(j\omega) = \frac{1}{1 + j\omega}$
 (C) $H(j\omega) = \frac{1}{j\omega}$ (D) $H(j\omega) = j\omega$

Direction for questions 24 and 25:

24. A signal $x(t)$ is multiplied with a rectangular pulse train $p(t)$ as shown. $x(t)$ can be recovered from product $x(t) \cdot p(t)$ by using an ideal low-pass filter, if $X(j\omega) = 0$ for



- (A) $\omega < 2000\pi$ (B) $\omega < 1000\pi$
 (C) $\omega > 1000\pi$ (D) $\omega > 2000\pi$
 25. From the above question, the pass band gain of the ideal low-pass filter needed to recover $x(t)$ from $x(t) \cdot p(t)$ is
 (A) 4 (B) 2 (C) 10 (D) 2.5

Practice Problems 2

Direction for questions 1 to 20: Select the correct alternative from the given choices.

1. Which of the following is not a Dirichlet condition?

- (A) $\frac{1}{T} \int_0^T |x(t)|^2 dt < \infty$
 (B) $\frac{1}{T} \int_0^T |x(t)|^2 dt > 0$
 (C) $x(t)$ has finite number of maxima and minima
 (D) $x(t)$ has a finite number of discontinuities

2. If $f(t)$ is an even function of time period $T/2$, then

- (A) $\int_{-T/2}^{T/2} f(t) dt = 2 \int_0^{T/2} f(t) dt$
 (B) $\int_0^T f(t) dt = \frac{1}{2} \int_0^{T/2} f(t) dt$
 (C) $\int_{-T/2}^{T/2} f(t) dt = 0$
 (D) A, B

3. $\int_{-T}^T f(t) dt = 0$, $f(t)$ is function of time period $2T$, then $f(t)$ is

- (A) Even function
 (B) Odd function
 (C) Symmetric function
 (D) Cannot say

4. The phase spectrum of a function is

- (A) Even function
 (B) Symmetric function
 (C) Anti symmetric function
 (D) Continuous function

5. Power spectrum is symmetrical about.

- (A) Horizontal axis
 (B) Horizontal axis passing through origin
 (C) Vertical axis passing through origin
 (D) Vertical axis.

6. The Fourier transform of $u(t)$, unit step function

- (A) $\pi \delta(\omega) + \frac{j}{\omega}$ (B) $\pi \delta(\omega) - \frac{j}{\omega}$
 (C) $\pi \delta(\omega) - \frac{1}{j\omega}$ (D) $-\pi \delta(\omega) + \frac{1}{j\omega}$

7. Match the Fourier transforms of group (1), to group (2) where $F\{x(t)\} = X(\omega)$

	Group (1)	Group (2)
P	$x(t - t_0)$	1. $\frac{dX(\omega)}{d\omega}$
Q	$e^{j\omega_0 t} x(t)$	2. $X(\omega - \omega_0)$
R	$\frac{dx(t)}{dt}$	3. $e^{-j\omega_0 t} X(\omega)$
S	$(-jt) x(t)$	4. $j\omega X(\omega)$

- (A) P - 2, Q - 3, R - 1, S - 4
 (B) P - 2, Q - 3, R - 1, S - 4
 (C) P - 3, Q - 2, R - 1, S - 4
 (D) P - 3, Q - 2, R - 4, S - 1

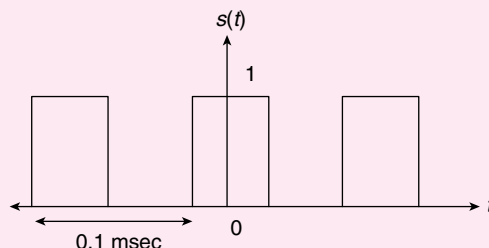
8. Determine the complex exponential Fourier series coefficients of $\sin^2 t$.

- (A) $\frac{-1}{4}, \frac{1}{2}, \frac{-1}{4}$ (B) $\frac{1}{2}, \frac{1}{4}, \frac{-1}{2}$
 (C) $1, \frac{1}{2}, -1$ (D) None

9. Fourier series of full-wave rectified sine wave, $\sin \omega_0 t$ for $t = 0$ to $T/2$ is
- (A) $\frac{2}{\pi} - \frac{4}{\pi} \left[\frac{\cos 2\omega_0 t}{3} + \frac{\cos 4\omega_0 t}{15} + \frac{\cos 6\omega_0 t}{35} + \dots \right]$
- (B) $\frac{2}{\pi} - \frac{4}{\pi} \left[\frac{\cos 3\omega_0 t}{3} - \frac{\cos 5\omega_0 t}{5} + \frac{\cos 7\omega_0 t}{7} + \dots \right]$
- (C) $\frac{2}{\pi} + \frac{4}{\pi} \left[\frac{\cos 2\omega_0 t}{3} - \frac{\cos 4\omega_0 t}{15} + \frac{\cos 6\omega_0 t}{35} + \dots \right]$
- (D) $\frac{2}{\pi} - \frac{4}{\pi} \left[\frac{\sin 2\omega_0 t}{3} + \frac{\sin 4\omega_0 t}{15} + \frac{\sin 6\omega_0 t}{35} + \dots \right]$
10. Fourier transform of sinusoidal signal $\sin \omega_0 t$ is
- (A) $\pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$
- (B) $\frac{\omega}{j} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
- (C) $j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$
- (D) $j\pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$
11. Which of the following signal will not have sine coefficients in Fourier series expansion?
- (A) $x(t) = 1 - t$ (B) $x(t) = \cos t$
- (C) $x(t) = \frac{1}{1+t}$ (D) $x(t) = (1 - t)^2$
12. The Fourier transform of the signal $e^{-3|t|}$ is
- (A) $\frac{6}{9 - \omega^2}$ (B) $\frac{6}{9 + \omega^2}$
- (C) $\frac{6}{\omega^2 - 9}$ (D) $\frac{-6}{\omega^2 + 9}$
13. Find the Fourier transform of $e^{-3|t-t_0|} + e^{3|t+t_0|}$
- (A) $\frac{12 \cos \omega t_0}{9 + \omega^2}$ (B) $\frac{12 \sin \omega t_0}{9 + \omega^2}$
- (C) $\frac{j12 \sin \omega t_0}{9 - \omega^2}$ (D) $\frac{12 \cos \omega t_0}{9 - \omega^2}$
14. Fourier transform of a Gaussian pulse is
- (A) sinc pulse (B) Sin function
- (C) Gaussian pulse (D) Impulse train
15. The average power of $x(n)$ in terms of Fourier series coefficients C_k is
- (A) $\sum_{k=0}^{\infty} |C_k|^2$ (B) $\frac{1}{N} \sum_{k=0}^{\infty} C_k^2$
- (C) $\frac{1}{N} \sum_{k=0}^{N-1} |C_k|^2$ (D) $\sum_{k=0}^{N-1} |C_k|^2$
16. If we replace 'z' in z transform of $x(n)$ with $e^{j\omega}$ we will get
- (A) continuous-time Fourier transform
- (B) discrete-time Fourier transform
- (C) Hilbert transform
- (D) complex z transform
17. The function $x(t)$ has the Fourier transform $X(\omega)$ then,
- $$\int_{-\infty}^{\infty} X(t) e^{-j\omega t} dt = ?$$
- (A) $\frac{1}{2\pi} X(\omega)$ (B) $2\pi X(-\omega)$
- (C) $2\pi x(-\omega)$ (D) $\frac{1}{2\pi} X(-\omega)$
18. If $F\{x(t)\} = \frac{\omega}{s^2 + \omega^2}$, then the value of $\lim_{t \rightarrow \infty} x(t)$ is
- (A) Cannot be determined (B) zero
- (C) unity (D) infinite
19. Match the time domain signals and their Fourier transforms
- | | List-I | List-II |
|----|--------------------------|---|
| P. | $1/2\pi$ | 1. $\delta(\omega - \omega_0)$ |
| Q. | $e^{j\omega_0 t} / 2\pi$ | 2. $\pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)]$ |
| R. | $\cos \omega_0 t$ | 3. $\delta(\omega)$ |
| S. | $\sin \omega_0 t$ | 4. $j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$ |
- (A) P - 1, Q - 3, R - 4, S - 2
- (B) P - 3, Q - 1, R - 4, S - 2
- (C) P - 3, Q - 1, R - 2, S - 4
- (D) P - 1, Q - 3, R - 2, S - 4
20. Delaying a signal by t_0 seconds in time-domain does
- (A) not change its amplitude and phase spectrum
- (B) not change its amplitude spectrum but the phase spectrum is changed by $-\omega t_0$
- (C) not change its phase spectrum but the amplitude spectrum is scaled by t_0
- (D) change phase spectrum by ωt_0 and amplitude spectrum by t_0 .

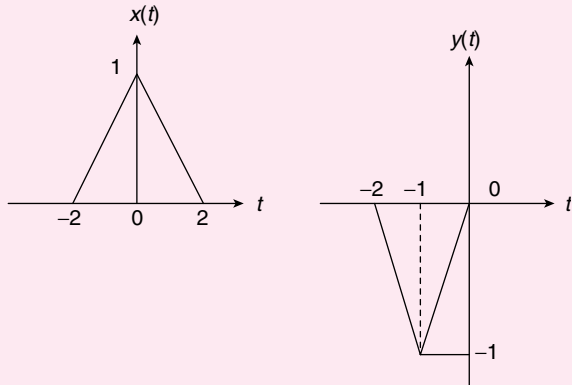
PREVIOUS YEARS' QUESTIONS

1. The Fourier transform of a conjugate symmetric function is always [2004]
- (A) Imaginary
- (B) Conjugate antisymmetric
- (C) Real
- (D) Conjugate symmetric
2. A rectangular pulse train $s(t)$ as shown in figure is convolved with the signal $\cos^2(4\pi \times 10^3 t)$, the convolved signal will be a [2004]



- (A) D.C. (B) 12 kHz sinusoid
(C) 8 kHz sinusoid (D) 14 kHz sinusoid

3. Let $x(t)$ and $y(t)$ (with Fourier transforms $X(f)$ and $Y(f)$, respectively) be related as shown in figure. [2004]



Then $Y(f)$ is

- (A) $-\frac{1}{2}X\left(\frac{f}{2}\right)e^{-j2\pi f}$ (B) $-\frac{1}{2}X\left(\frac{f}{2}\right)e^{j2\pi f}$
(C) $-X\left(\frac{f}{2}\right)e^{j2\pi f}$ (D) $-X\left(\frac{f}{2}\right)e^{-j2\pi f}$

4. The output $y(t)$ of a linear time invariant system is related to its input $x(t)$ by the following equation: $y(t) = 0.5 \times (t - t_d + T) + X(t - t_d) + 0.5x(t - t_d - T)$. The filter transfer function $H(\omega)$ of such a system is given by [2005]

- (A) $(1 + \cos\omega T)e^{-j\omega t_d}$
(B) $(1 + 0.5 \cos\omega T)e^{-j\omega t_d}$
(C) $(1 + \cos\omega T)e^{-j\omega t_d}$
(D) $(1 + 0.5 \cos\omega T)e^{-j\omega t_d}$

5. For a signal $x(t)$, the Fourier transform is $X(f)$. Then, the inverse Fourier transform of $X(3f + 2)$ is given by [2005]

- (A) $\frac{1}{2}x(1/2)e^{j3\pi t}$ (B) $\frac{1}{3}x(t/3)e^{\frac{-j4\pi t}{3}}$
(C) $3X(3t)e^{-j4\pi t}$ (D) $X(3t + 2)$

6. A low-pass filter having a frequency response $H(j\omega) = A(\omega)e^{j\phi(\omega)}$ does not produce any phase distortion if [2006]

- (A) $A(\omega) = C\omega^2$, $\phi(\omega) = K\omega^3$
(B) $A(\omega) = C\omega^2$, $\phi(\omega) = K\omega$
(C) $A(\omega) = C\omega$, $\phi(\omega) = K\omega^2$
(D) $A(\omega) = C$, $\phi(\omega) = K\omega^{-1}$

7. Let $x(t) \leftrightarrow X(j\omega)$ be Fourier transform pair. The Fourier transform of the signal $x(5t - 3)$ in terms of $X(j\omega)$ is given as [2006]

- (A) $\frac{1}{5}e^{\frac{-j3\omega}{5}}X\left(\frac{j\omega}{5}\right)$ (B) $\frac{1}{5}e^{\frac{j3\omega}{5}}X\left(\frac{j\omega}{5}\right)$
(C) $\frac{1}{5}e^{-j3\omega}X\left(\frac{j\omega}{5}\right)$ (D) $\frac{1}{5}e^{j3\omega}X\left(\frac{j\omega}{5}\right)$

8. A signal $m(t)$ with band width 500 Hz is first multiplied by a signal $g(t)$ where

$$g(t) = \sum_{K=-\infty}^{\infty} (-1)^K \delta(t - 0.5 \times 10^{-4} K)$$

The resulting signal is then passed through an ideal low-pass filter with band width 1 kHz. The output of the low-pass filter would be: [2006]

- (A) $\delta(t)$ (B) $m(t)$
(C) 0 (D) $m(t)\delta(t)$

9. The signal $x(t)$ is described by

$$x(t) = \begin{cases} 1 & \text{for } -1 \leq t \leq +1 \\ 0 & \text{otherwise} \end{cases}$$

Two of the angular frequencies at which its Fourier transform becomes zero are [2008]

- (A) $\pi, 2\pi$ (B) $0.5\pi, 1.5\pi$
(C) $0, \pi$ (D) $2\pi, 2.5\pi$

10. The Fourier series of a real periodic function has only P. Cosine terms if it is even

Q. Sine terms if it is even

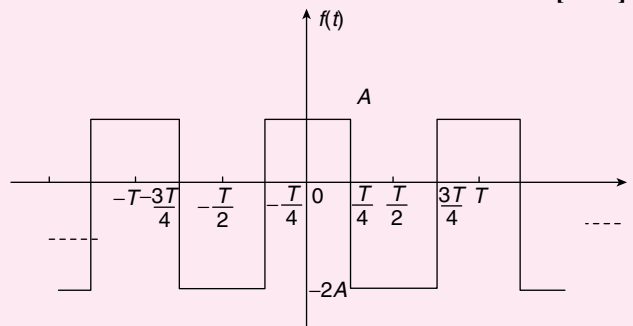
R. Cosine terms if it is odd

S. Sine terms if it is odd

Which of the above statements are correct? [2009]

- (A) P and S (B) P and R
(C) Q and S (D) Q and R

11. The trigonometric Fourier series for the waveform $f(t)$ shown below contains [2010]

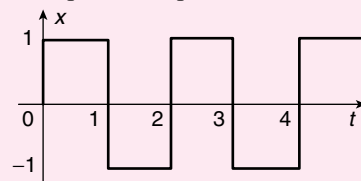


- (A) only cosine terms and zero value for the d.c. component
(B) only cosine terms and a positive value for the d.c. component
(C) only cosine terms and a negative value for the d.c. component
(D) only sine terms and a negative for the d.c. component

12. The Fourier transform of a signal $h(t)$ is $H(j\omega) = (2\cos \omega)(\sin 2\omega)/\omega$. The value of $h(0)$ is [2012]

- (A) 1/4 (B) 1/2 (C) 1 (D) 2

13. Consider the periodic square wave in the figure shown.



The ratio of the power in the 7th harmonic to the power in the 5th harmonic for this waveform is closest in value to _____. [2014]

14. The value of the integral $\int_{-\infty}^{\infty} \sin^2(5t) dt$ is _____. [2014]

15. A real-valued signal $x(t)$ limited to the frequency band $|f| \leq \frac{W}{2}$ is passed through a linear time invariant system whose frequency response is

$$H(f) = \begin{cases} e^{-j4\pi f}, & |f| \leq \frac{W}{2} \\ 0, & |f| > \frac{W}{2} \end{cases}$$

The output of the system is [2014]

- (A) $x(t+4)$ (B) $x(t-4)$
(C) $x(t+2)$ (D) $x(t-2)$

16. Which one of the following is an eigen function of the class of all continuous time, linear, time invariant systems ($u(t)$ denotes the unit step function)? [2016]

- (A) $e^{j\omega_0 t} u(t)$ (B) $\cos(\omega_0 t)$
(C) $e^{j\omega_0 t}$ (D) $\sin(\omega_0 t)$

17. A continuous time sinusoid of frequency 33Hz is multiplied with a periodic Dirac impulse train of frequency 46Hz. The resulting signal is passed through an ideal analog low pass filter with a cutoff frequency of 23Hz. The fundamental frequency (in Hz) of the output is _____. [2016]

18. A network consisting of a finite number of linear resistor (R), inductor (L) and capacitor (C) elements, connected all in series or all in parallel, is excited with a source of the form.

$$\sum_{k=1}^3 a_k \cos(k\omega_0 t) \text{ where } a_k \neq 0, \omega_0 \neq 0$$

The source has nonzero impedance. Which one of the following is a possible form of the output measured across a resistor in the network? [2016]

(A) $\sum_{k=1}^3 b_k \cos(k\omega_0 t + \phi_k)$ where $b_k \neq a_k, \forall k$

(B) $\sum_{k=1}^4 b_k \cos(k\omega_0 t + \phi_k)$ where $b_k \neq 0, \forall k$

(C) $\sum_{k=1}^3 a_k \cos(k\omega_0 t + \phi_k)$

(D) $\sum_{k=1}^2 a_k \cos(k\omega_0 t + \phi_k)$

19. Consider the signal $x(t) = \cos(6\pi t) + \sin(8\pi t)$, where t is in seconds. The Nyquist sampling rate (in samples/second) for the signal $y(t) = x(2t+5)$ is [2016]

- (A) 8 (B) 12
(C) 16 (D) 32

20. If the signal $x(t) = \frac{\sin(t)}{\pi t} * \frac{\sin(t)}{\pi t}$ with $*$ denoting the convolution operation, then $x(t)$ is equal to [2016]

- (A) $\frac{\sin(t)}{\pi t}$ (B) $\frac{\sin(2t)}{2\pi t}$
(C) $\frac{2\sin(t)}{\pi t}$ (D) $\left(\frac{\sin(t)}{\pi t}\right)^2$

21. A signal $2\cos\left(\frac{2\pi}{3}t\right) - \cos(\pi t)$ is the input to an LTI

system with the transfer function $H(s) = e^s + e^{-s}$.

If C_k denotes the k^{th} coefficient in the exponential Fourier series of the output signal, then C_3 is equal to [2016]

- (A) 0 (B) 1
(C) 2 (D) 3

ANSWER KEYS

EXERCISES

Practice Problems I

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. B | 2. C | 3. B | 4. A | 5. A | 6. C | 7. D | 8. B | 9. B | 10. A |
| 11. B | 12. C | 13. C | 14. D | 15. D | 16. B | 17. A | 18. C | 19. D | 20. A |
| 21. A | 22. D | 23. B | 24. B | 25. D | | | | | |

Practice Problems 2

- | | | | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| 1. B | 2. A | 3. B | 4. C | 5. C | 6. B | 7. D | 8. A | 9. A | 10. C |
| 11. B | 12. B | 13. A | 14. C | 15. D | 16. B | 17. C | 18. A | 19. C | 20. B |

Previous Years' Questions

- | | | | | | | | | | |
|-------|-------|-----------------|------------------|-------|-------|--------|-------|------|-------|
| 1. C | 2. A | 3. B | 4. A | 5. B | 6. B | 7. A | 8. B | 9. A | 10. A |
| 11. C | 12. C | 13. 0.5 to 0.52 | 14. 0.19 to 0.21 | 15. D | 16. C | 17. 13 | 18. A | | |
| 19. C | 20. A | 21. B | | | | | | | |